

BUTLER ISLAND, Antarctica

WMO: 892660

Lat: 72.206S

Lon: 60.170W

Elev: 115

StdP: 99.95

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -36.4	(c) -34.6	(d) -40.5	(e) 0.1	(f) -36.6	(g) -38.7	(h) 0.1	(i) -35.1	(j) 18.4	(k) -25.3	(l) 16.1	(m) -26.2	(n) 3.2	(o) 190	(p) 1.225

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 5.5	(c) 4.9	(d) 1.6	(e) 3.0	(f) 0.2	(g) 1.2	(h) -1.2	(i) 0.6	(j) 3.0	(k) -0.7	(l) 0.9	(m) -1.8	(n) -0.6	(o) 2.6	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-1.5	3.4	0.3	-2.4	3.1	-0.6	-3.2	2.9	-1.5	10.9	3.0	8.6	1.0	6.7	-0.6	5.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.2	(b) 13.5	(c) 11.4	(d) DB	(e) -37.8	(f) 10.1	(g) 2.3	(h) 3.2	(i) -39.5	(j) 12.5	(k) -40.9	(l) 14.4	(m) -42.2	(n) 16.2	(o) -43.9	(p) 18.5	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.3	-3.3	-8.3	-16.0	-21.9	-23.1	-23.5	-24.8	-23.8	-20.9	-17.3	-8.9	-3.7
	DBStd	9.59	3.01	4.43	5.97	5.30	6.13	6.73	6.83	6.98	6.99	5.67	3.97	2.88
	HDD10.0	9606	411	512	807	956	1025	1004	1077	1047	927	847	567	424
	HDD18.3	12647	669	746	1065	1206	1283	1254	1336	1305	1177	1106	817	683
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	4.3	3.7	4.3	4.2	3.9	3.9	4.1	4.4	4.3	4.9	5.0	4.6	4.4
(15) Precipitation	PrecAvg	655	38	46	50	49	58	48	58	66	72	71	60	39
	PrecMax	872	67	63	109	105	107	84	151	108	113	137	101	80
	PrecMin	441	12	21	17	18	23	20	28	21	44	27	25	13
	PrecStd	104	14	12	19	20	20	17	26	18	18	27	21	17
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	6.3	4.9	-0.1	1.4	1.1	1.1	-0.4	1.7	3.2	5.0	6.7
		MCWB	2.8	2.8	1.7	-1.7	-1.6	-1.5	-1.5	-3.1	-1.2	0.1	0.8	2.4
	2%	DB	4.6	3.3	1.1	-6.9	-3.2	-3.5	-3.4	-3.5	-1.5	-1.7	1.3	3.8
		MCWB	1.0	0.5	-1.7	-8.1	-5.3	-5.3	-5.3	-5.9	-4.0	-4.4	-1.6	0.7
	5%	DB	2.3	0.1	-3.9	-11.3	-9.0	-8.6	-8.4	-6.6	-4.9	-6.0	-1.8	1.7
		MCWB	-0.3	-2.2	-5.5	-11.7	-10.2	-9.6	-9.6	-8.3	-6.9	-6.8	-3.5	-0.7
	10%	DB	0.4	-2.3	-8.0	-14.7	-14.7	-14.3	-15.4	-13.9	-10.3	-9.8	-3.5	0.0
		MCWB	-1.2	-3.3	-8.4	-15.2	-15.0	-14.8	-15.8	-14.4	-11.2	-10.4	-4.7	-1.5
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.8	2.0	-4.8	-6.1	-3.1	-1.4	-2.7	-2.4	-4.1	-0.4	1.3
		MCDB	4.6	6.0	5.0	-2.6	-4.1	-1.0	0.9	-0.3	0.3	-2.1	1.8	4.0
	2%	WB	0.0	0.3	-1.9	-8.6	-9.7	-7.8	-4.8	-4.7	-5.0	-6.8	-2.4	-0.3
		MCDB	2.1	2.8	0.6	-7.8	-8.2	-6.4	-2.7	-2.6	-2.6	-5.7	-0.4	1.1
	5%	WB	-1.0	-2.2	-5.3	-11.4	-14.1	-12.6	-8.5	-7.0	-8.7	-9.9	-4.0	-1.3
		MCDB	0.2	-0.7	-3.6	-10.9	-13.9	-11.9	-7.2	-4.9	-7.2	-9.1	-3.0	-0.2
	10%	WB	-2.1	-3.9	-8.4	-14.6	-17.1	-16.5	-13.4	-11.1	-13.4	-12.2	-5.3	-2.3
		MCDB	-1.4	-3.4	-7.9	-14.2	-16.8	-16.1	-12.6	-10.1	-12.8	-11.8	-4.5	-1.4
(35) Mean Daily Temperature Range	MDBR		5.5	5.7	5.8	5.4	5.8	5.9	6.7	6.5	6.9	6.5	6.0	5.5
	5% DB	MCDBR	9.4	9.4	12.4	11.9	16.0	14.4	15.3	15.5	14.7	13.3	9.7	9.5
		MCWBR	5.6	7.8	9.3	9.9	14.3	12.8	12.9	12.8	12.2	10.5	7.2	6.5
	5% WB	MCDBR	6.5	8.0	11.3	11.3	10.5	11.7	14.6	14.7	14.3	10.9	7.1	6.4
		MCWBR	5.1	6.6	9.3	10.2	9.7	10.9	13.0	12.9	12.3	9.7	6.1	5.2
(40) Clear-Sky Solar Irradiance	taub		0.256	0.242	0.211	0.138	N/A	N/A	N/A	0.140	0.208	0.228	0.241	0.255
	taud		2.242	2.266	2.261	1.987	N/A	N/A	N/A	1.934	2.201	2.226	2.246	2.233
	Ebn at Noon		974	919	823	551	0	0	0	512	805	932	990	994
	Edh at Noon		109	89	61	25	0	0	0	26	63	92	108	116
(44) All-Sky Solar Radiation	RadAvg		7.82	5.07	2.27	0.47	0.00	0.00	0.00	0.19	1.48	4.06	7.09	8.87
	RadStd		0.23	0.23	0.07	0.03	0.00	0.00	0.00	0.01	0.05	0.11	0.14	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page